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CPSC 320 A2

1 Slow as a logarithm

expanding on $\log_b(n) \in O(n^x)$ with the definition of big O,

$$\exists c \in \mathbb{R}, \exists n_0 \in \mathbb{R}, c > 0$$
$$\forall n \geq n_0 > 0,$$
$$\log_b(n) \leq c \cdot n^x$$

for every $b > 1$ and $x > 0$
let $n_0 = 1$ and $c = 1$

$$\log_b(1) \leq 1 \cdot 1^x$$
$$0 \leq 1 \cdot 1^x$$
$$0 \leq 1$$

so $n_0 = 1$ and $c = 1$ is an valid choosing of c and n_0

and we want, by the definition of little o

$$\lim_{n \to \infty} \left(\frac{\log_b(n)}{n^x} \right) = 0$$

Question 2: I wanna be the very best

a)

this is a combination question.

Question 3

3a)

Choose $c = 1, n_0 = 1$

$$\begin{aligned}
\ln(n!) &= \ln(n \cdot (n-1)!) \\
&= \ln(n) + \ln((n-1)!) \\
&= \ln(n) + \ln(n-1) + ... + \ln(1) && \text{there are } n \text{ terms} \\
&\leq \ln(n) + \ln(n) + ... + \ln(n) && \text{since } \ln(n) \text{ is an increasing function} \\
&= n \cdot \ln(n) \\
&\leq c \cdot n \ln(n)
\end{aligned}$$

therefore $\exists c \in \mathbb{R}^+, \exists n_0 \in \mathbb{N}, \forall n \geq n_0, \ln(n!) \leq c \cdot n \ln(n)$

which is the definition of $\ln(n!) \in O(n \ln n)$

3b

proving the hint first:
$\forall n \in \mathbb{N}, n^n \geq (n+1)^{n-1}$

base case $n = 1$

$$1^1 \geq (1+1)^{1-1}$$
$$1 \geq 2^0$$
$$1 \geq 1$$

Assume for inductive hypothesis

$$k^k \geq (k+1)^{k-1}$$

for $1 \leq k$ and $k \in \mathbb{N}$

from the inductive hypothesis

$$
\begin{aligned}
k^k &\geq (k+1)^{k-1} \\
k^k &\geq \frac{(k+1)^k}{k+1} \\
\frac{k^k}{(k+1)^k} &\geq \frac{1}{k+1} \\
\left(\frac{k}{k+1}\right)^k &\geq \frac{1}{k+1} \\
\left(\frac{k+1}{k+2}\right)^k > \left(\frac{k}{k+1}\right)^k &\geq \frac{1}{k+1} \ (*) \\
\left(\frac{k+1}{k+2}\right)^k &\geq \frac{1}{k+1} \\
\frac{(k+1)^k}{(k+2)^k} &\geq \frac{1}{k+1} \\
(k+1)^k &\geq \left(\frac{(k+2)^k}{k+1}\right) \\
(k+1)^k(k+1) &\geq (k+2)^k \\
(k+1)^{k+1} &\geq (k+2)^k \\
(k+1)^{k+1} &\geq (k+1+1)^{k+1-1}
\end{aligned}
$$

to explain (*), if we put $\frac{k+1}{k+2}$ and $\frac{k}{k+1}$ into a ratio, we end up with

$$\frac{k^2+2k+1}{k^2+2k}$$

show casing that $\frac{k+1}{k+2}$ is strictly greater than $\frac{k}{k+1}$

therefore by mathematical induction, we have

$$\forall n \in \mathbb{N}, n^n \geq (n+1)^{n-1}$$

Next, proving $(n!)^2 \geq n^n, \forall n \in \mathbb{N}$

base case, $n = 1$

$$(1!)^2 = 1 \geq 1 = 1^1$$

For inductive hypothesis assume that $(k!)^2 \geq k^k, \forall k \in \mathbb{N}, k \geq 1$

$$
\begin{aligned}
((k+1)!)^2 &= ((k+1)k!)^2 \\
&= (k+1)^2(k!)^2 \\
&\geq (k+1)^2 \cdot k^k && \text{by the inductive hypothesis} \\
&= (k+1) \cdot (k+1) \cdot k^k \\
&\geq (k+1) \cdot (k+1) \cdot (k+1)^{k-1} && \text{by using the hint} \\
&= (k+1) \cdot (k+1)^k \\
&= (k+1)^{k+1}
\end{aligned}
$$

therefore by mathematical induction we have

$$(n!)^2 \geq n^n, \forall n \in \mathbb{N}$$

■

3c

from part a, we have $\ln(n!) \in O(n\ln(n))$

we now seek to prove $\ln(n!) \in \Omega(n\ln(n))$

choose $c = \frac{1}{2}$ and $n_0 = 1$,

$$\begin{aligned}
(n!)^2 &\geq n^n && \text{from part b} \\
\ln(n!^2) &\geq \ln(n^n) && \text{since } \ln \text{ is a increasing function and } n \geq n_0 \geq 1 \\
2\ln(n!) &\geq \ln(n^n) \\
\ln(n!) &\geq \frac{1}{2}\ln(n^n) && \text{divide both sides by 2}
\end{aligned}$$

then we get

$$\begin{aligned}
\ln(n!) &\geq \frac{1}{2}n\ln(n) \\
&= cn\ln(n)
\end{aligned}$$

therefore $\exists c \in \mathbb{R}^+, \exists n_0 \in \mathbb{N}, \forall n \geq n_0, c \cdot n\ln(n) \leq \ln(n!)$

which is the definition of $\ln(n!) \in \Omega(n\ln(n))$

and since we have both $\ln(n!) \in \Omega(n\ln(n))$ and $\ln(n!) \in O(n\ln(n))$

we can conclude $\ln(n!) \in \Theta(n\ln(n))$

Question 4

`H` is home `A` is school A `B` is school B `W` is work

a)

3 times for symmetric

6 times for asymmetric

b)

2 times for symmetric, because running from (home, A) gives implicitly (home, B), and we just need to run it again from (A,B)

3 times for asymmetric.

we need all 6,

```
(H, A)
(A, H)
(A, B)
(B, A)
(H, B)
(B, H)
```

running noodle maps on `(H, A)` gives also `(H, B)`

again, on `(A, B)` gives `(A, H)`

again on `(B, H)` gives `(B, A)`

thus, getting all 6 in 3 runs

c)

if We are working only under part a with part c,

5 times if symmetric.

```
(H, A)
(H, B)
(A, B)
(A, W)
(B, W)
```

if asymmetric, it's 6 times, one per each edge

```
(H, A)
(H, B)
(A, B)
(B, A)
(A, W)
(B, W)
```

now working under part b

2 times for symmetric. in this case we need four edges,

```
(H, A)
(B, A)
(B, W)
(A, W)
```

run it on `(B, A)` gives `(B, W)`

run it again on `(A, W)` gives `(H, A)` as well.

in the asymmetric case, it's 3 times

```
(H, A)
(H, B)

(A, W)
(A, B)

(B, W)
(B, A)
```

run it on `(A, W)` gives `(A, B)`

run it on `(B, A)` gives, `(B, W)`

run it on `(H, A)` gives `(H, B)`